Summary in Graph

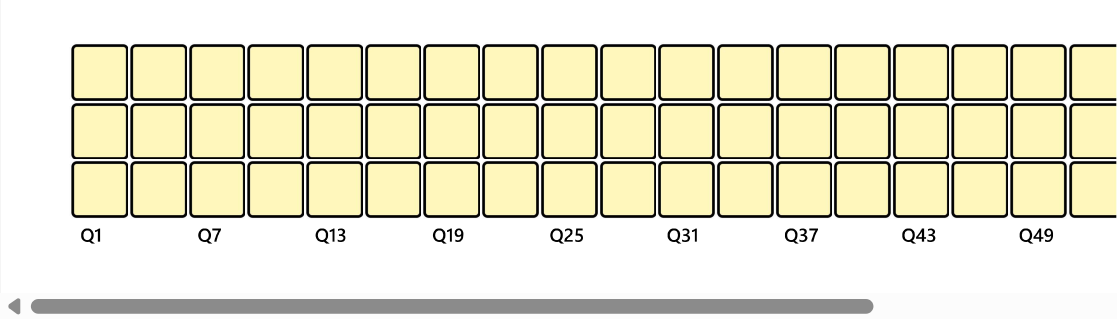
Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (51) out of 508 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

I bought a new plant for my garden. Anika said it was a red rose, Bill said it was a purple daisy, and Cathy said it was a red dahlia. Each person was correct in stating either the colour or the type of plant. What was the plant that I bought?

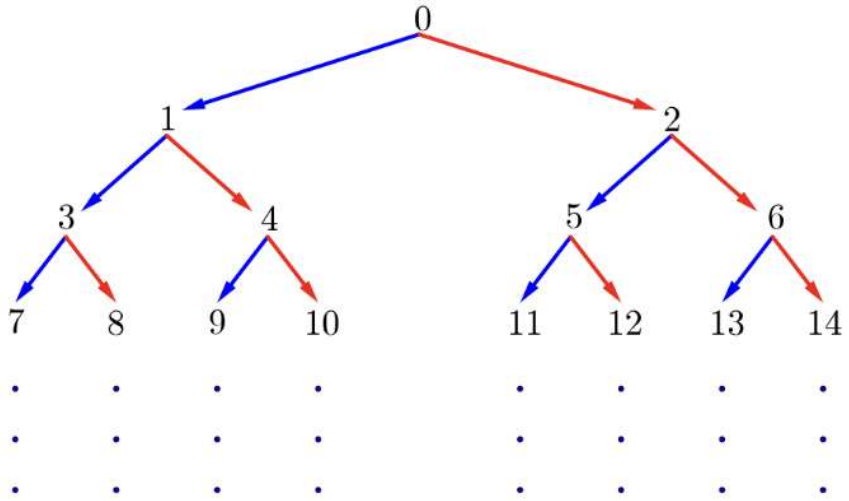
A. purple dahlia
B. purple rose

C. red dahlia
D. red daisy

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

Consider the number tree shown to the right. In this number tree, the integers greater than or equal to 0 are written out in increasing order, with the top row containing one integer and every row after containing twice as many integers as the row above it. If row 1 contains the integer 0 , what is the fifth number (from left) in row 10?



A. 512
B. 513
C. 514
D. 515

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

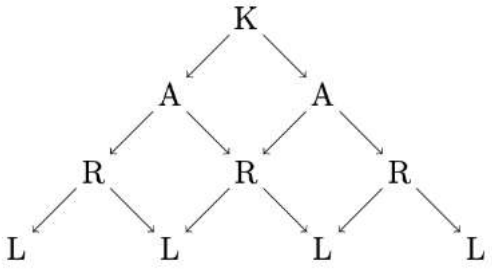
The average of a, b and c is 16 . The average of c, d and e is 26 . The average of $a, b, c, d,$ and e is 20 . The value of c is -

A. 10
B. 20
C. 21
D. 26

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

In the diagram, how many paths can be taken to spell "KARL"?



A. 4
B. 16
C. 6
D. 8

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Choose the correct answer:
Judy___watching horror films although she has nightmares afterwards.

A. like
B. likes
C. liked
D. will like

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 04sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

There are 5 empty glasses on a table. One is facing down and four are facing up.



Flipping a glass changes it from facing up to facing down, or from facing down to facing up. In one turn, you must flip exactly three different glasses. The glasses which are flipped do not need to be adjacent.

A. 2 turns
B. 3 turns
C. 5 turns
D. it is not possible to make all glasses facing up

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

Chris lies on Fridays, Saturdays and Sundays, but he tells the truth on all other days. Mark lies on Tuesdays, Wednesdays and Thursdays, but he tells the truth on all other days. On what day of the week would they both say: "Tomorrow I will lie."?

A. Monday
B. Thursday

C. Friday
D. Sunday

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Last Thursday, each of the students in M. Fermat's class brought one piece of fruit to school. Each brought an apple, a banana, or an orange. In total, 20% of the students brought an apple and 35% brought a banana. If 9 students brought oranges, how many students were in the class?

A. 18
B. 64
C. 24
D. 20

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

The organizer for a sports league with four teams has entered some of the end-of-season data into the table shown. Each team played 27 games and each game resulted in a win for one team and a loss for the other team, or in a tie for both teams. Each team earned 2 points for a win, 0 points for a loss, and 1 point for a tie.

Team Name	Games Played	Number of Wins	Number of Losses	Number of Ties	Total Points
P	27	10	14		23
Q	27				
R	27				25
S	27				

At the end of the season, Team S had 4 more wins than losses. What are the total points of Team S after finishing the season?

A. 28
B. 30
C. 31
D. 34

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

The store ___ by hundreds of people everyday, I don't understand why my favorite book ___ yet.

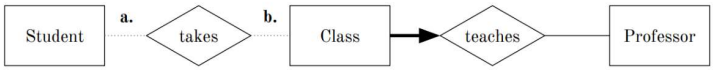
A. visited / hasn't been bought
B. is visited / hasn't been bought
C. was visited / hadn't bought
D. is visited / hadn't been bought

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Complete the following ER diagram to enforce that every student must be enrolled in at least one class and that classes can have any number of students.



A. (a) is a Bold Line without an arrow, (b) is a Bold Line without an arrow.
B. (a) is a Bold Line with an arrow, (b) is a Bold Line without an arrow.
C. (a) is a Bold Line without an arrow, (b) is a thin Line without an arrow.
D. (a) is a thin Line without an arrow, (b) is a bold Line without an arrow.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Suppose we have a relation $\mathrm{R}(\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F})$ with functional dependencies(FDs):

$$\mathrm{A} \rightarrow \mathrm{B}$$
$$\mathrm{BCD} \rightarrow \mathrm{E}$$
$$\mathrm{E} \rightarrow \mathrm{F}$$

Suppose there are at most 2 different values for each of attributes A, C, and D. What is the maximum number of different values for attribute F?

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider a $\mathrm{B}+$ tree defined on a database relation. The $\mathrm{B}+$ tree's leaves hold RIDs(record pointers), not the actual tuples. The size of the index key values is such that N (key, pointer) or (key, RID) pairs can be stored in each block of the index.

What is the maximum relation size (i.e., the maximum number of tuples) that can be indexed using a $\mathrm{B}+$ tree of height 3? (Height of a single node $\mathrm{B}+$ tree is considered as 1)

A. $\theta(\mathrm{N})$
B. $\theta(\mathrm{N}^2)$
C. $\theta(\mathrm{N}^3)$
D. $\theta(\mathrm{N}^4)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Choose all statements which are guaranteed to be true if $f = n(\sin(n) + 2)$

A. $f = \Omega(1)$
B. $f = \Theta(n)$
C. $f = O\left(n^2\right)$
D. None of the other options are true

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider 4 characters a, b, c and d. Below are the corresponding codes to encode these characters.

$$a = < 00 >, b = < 01 >, c = < 11 >, d = < 101 >.$$

Which of the following option is true about given statements?

- $\mathrm{S1}$: This is a prefix-free code.
- $\mathrm{S2}$: This code could be generated by Huffman's algorithm.

A. $\mathrm{S1}$ is true but $\mathrm{S2}$ is not true
B. $\mathrm{S1}$ is not true but $\mathrm{S2}$ is true
C. Both are true
D. Both are false

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

You are given two sorted lists: one with m elements and the other with n elements. When merging these two lists into a single sorted list, how many comparisons are required in the worst-case and best-case scenarios, respectively?

A. $\mathrm{m} + \mathrm{n}$ and min(m, n)
B. $m + n - 1$ and max(m, n)
C. $\max(m, n)$ and $\min(m, n)$
D. $m + n - 1$ and min(m, n)

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

A self-printing queue is a queue of integers, implemented as a singly linked list, which prints the contents of the queue to standard output after every three (enqueue or dequeue) operations. For instance, the sequence enqueue(0), dequeue(), enqueue(0) prints 0.

What does the following sequence of operations on an initially empty self-printing queue print?
enqueue(0), enqueue(1), dequeue(), enqueue(2), enqueue(3), dequeue(), enqueue(4), enqueue(5), dequeue(), enqueue(6), enqueue(7), dequeue()

A. 1 2 3 4 5 6 6 7 7
B. 1 2 2 3 3 4 4 5 5 6
C. 1 2 3 3 4 5 4 5 6 7
D. 0 2 0 2 4 5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Inorder traversing a tree resulted E A C K F H D B G; the preorder traversal would return.

A. F A E K C D B H G
B. F A E K C D H G B
C. E A F K H D C B G
D. F E A K D C H B G

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following Python code

```
siblings_ids = (1, 2, [3, 4], 11, 9)
sibling_id[2].append(5)
siblings_ids[4] = 10
print(sibling_ids)
```

What is the output after running the above piece of code?

A. $(1, 2, [3, 4, 5], 11, 10)$
B. $(1, 2, (3, 4, 5), 11, 9)$
C. $(1, 2, (3, 4, 5), 11, 9, 10)$
D. TypeError: 'tuple' object does not support item assignment

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider following python code -

```
c = {}
c[1] = 1
c['1'] = 2
c[1] += 1
print(c)
```

What is the output of the above code?

A. $\{1 : 1, '1' : 2\}$
B. $\{1 : 2, '1' : 2\}$
C. $\{1 : 1, '1' : 3\}$
D. $\{1 : 2, '1' : 3\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the subspace W of R^8, defined as:

$W = \{(y_1, y_2, \ldots, y_8) \in R^8 : y_n = 2y_{n-1} + y_{n-2} \text{ for } 3 \leq n \leq 8\}$.

What is the dimension of W?

A. 2
B. 3
C. 6
D. 8

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Urn A contains 5 white and 7 black balls. Urn B contains 3 white and 12 black balls. A fair coin is flipped:
- If the outcome is heads, a ball is selected from Urn A.
- If the outcome is tails, a ball is selected from Urn B.
Given that a black ball is selected, what is the probability that the coin landed tails?

A. $1/2$
B. $48/83$
C. $12/15$
D. $7/12$

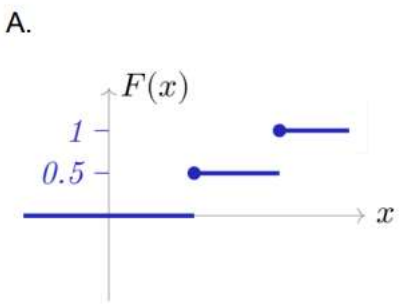
Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

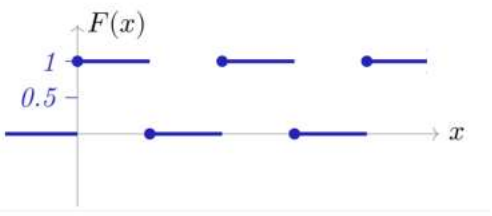
Which of the following is/are valid graphs for the cumulative distribution function (CDF) of a discrete random variable?

A.

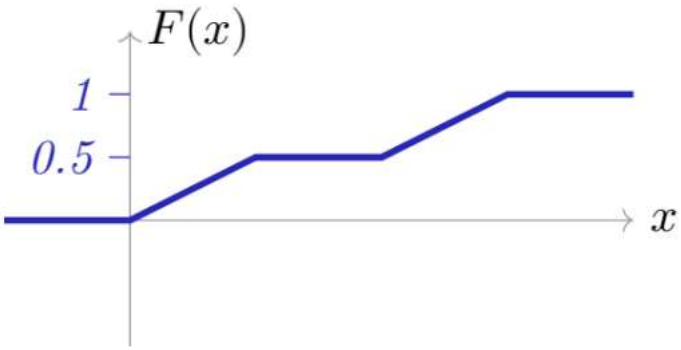
A.



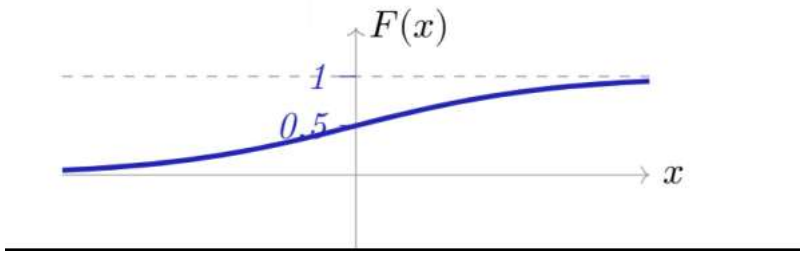
B.



C.



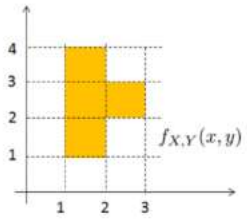
D.



Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Consider the joint PDF $f_{X,Y}(x,y)$ = is uniform and equal to $\frac{1}{4}$ over the yellow shaded regions in the diagram below. The marginal PDF of X, $f_X(x)$, is given as:



$$f_X(x) = \begin{cases} a, & \text{if } 1 < x \leq 2 \\ b, & \text{if } 2 < x \leq 3 \\ 0, & \text{otherwise.} \end{cases}$$

What are the values of a and b ?

A. $a = \frac{2}{4}, b = \frac{1}{4}$
B. $a = \frac{3}{4}, b = \frac{1}{4}$
C. $a = 1, b = \frac{2}{4}$
D. $a = \frac{1}{4}, b = 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

A random variable X has the following probability density function:

$$f(x) = \begin{cases} c\left(x + x^2\right), & x \in [0,1] \\ 0, & \text{otherwise} \end{cases}$$

What is the value of c ?

A. $c = \frac{5}{6}$
B. $c = \frac{6}{5}$

C. $c = \frac{2}{3}$
D. $c = \frac{3}{2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let A and B be two events. Suppose the probability that neither A nor B occurs is $2/3$. What is the probability that one or both events occur?

A. $1/3$
B. $2/3$
C. $1/2$
D. 1

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Gradient of a continuous and differentiable function

A. is zero at a minimum
B. is non-zero at a maximum
C. is zero at a saddle point
D. decreases as you get closer to the minimum

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

How many local minima does the function $f(x) = x^4 - \frac{4}{3}x^3 - 12x^2$ have?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Assume you have a single data observation k, what is the maximum likelihood estimate for θ given the following likelihood function:

$$\mathcal{L}(\theta) = \frac{\theta^k}{k!}e^{-\theta}$$

A. $1/k$
B. k
C. k/n
D. $k!$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following 1-dimensional samples from two classes:
- Class 1: $\{-2, 0, 2, 4\}$
- Class 2: $\{-1, 3, 6, 7\}$
A k-Nearest Neighbors (kNN) classifier is used with $\mathrm{k} = 2$, and in the case of a tie (one neighbor from each class), the classifier defaults to Class 1.
What will be the decision boundaries based on the given rule ?

A. Decision boundary at $x = 1.5$
B. Decision boundary at $x = 5.5$
C. Decision boundary at $x = 3.5$
D. Decision boundary at $x = -1.5$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

What is the correct computation for the entropy of a dataset with 10 positive examples and 23 negative examples?

A. $-(10/33) * \log(23/33) - (23/33) * \log(10/33)$
B. $-(10/33) * \log(10/33) - (23/33) * \log(23/33)$
C. $-(10/23) * \log(10/23) - (23/10) * \log(23/10)$
D. $(10/33)^* \log(10/33) + (23/33) * \log(23/33)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

In soft margin SVMs, the slack variable ξ_i are defined in the following constriant:

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

The slack variable ζ_i have to satisfy which of the following conditions?

A. ξ_i < 0
B. ξ_i = 0
C. ξ_i > 0
D. $\xi_i \geq$ 0

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

We define an evaluation function for a heuristic search problem as:
$f(n) = (w * g(n)) + ((1 - w) * h(n))$ where $g(n)$ is the cost of the best path found from the start state to state $\mathrm{n}, \mathrm{h}(\mathrm{n})$ is an admissible heuristic function that estimates the cost of a path from n to a goal state, and $0.0 \leq w \leq 1.0$. What search algorithm do you get when $w = 1.0$?

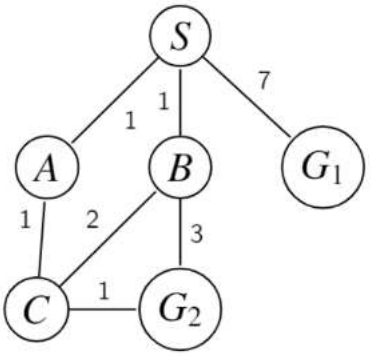
A. Breadth-First search
B. Uniform-Cost search

C. Greedy Best-First search
D. Algorithm A* search

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Which goal is reached and what is the total cost of the solution found for the following state-space graph when using Breadth-First Search and Uniform-Cost Search (S is the start state, G1 and G2 are the goal states, arcs are bidirectional, no repeated state checking, break any ties alphabetically)?



A. BFS: G1 (Cost: 7), UCS: G2 (Cost: 4)
B. BFS: G2 (Cost: 4), UCS: G1 (Cost: 7)
C. BFS: G2 (Cost: 4), UCS: G2 (Cost: 4)
D. BFS: G1 (Cost: 7), UCS: G2 (Cost: 3)

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Which of the following statements is/are TRUE ?

A. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $\sim Q \wedge \sim P$
B. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $Q \vee P$
C. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $Q \vee (P \wedge \sim Q)$
D. $(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$ is equal to $P \vee (Q \wedge \sim P)$.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following database schema:
Product (pid, name, brand, price, color), where **pid** is the primary key.
Consider the following SQL queries:

Query P :

```
SELECT    brand
FROM      Product
WHERE     color = 'green'
GROUP BY  brand
HAVING    COUNT(*) < 3
```

Query Q :

```
SELECT  DISTINCT brand
FROM    Product p
WHERE   (SELECT  COUNT(*)
         FROM    Product q
         WHERE   q.brand = p.brand
                 AND q.color = 'green') < 3
```

Which of the following is correct?

A. Query P, Q are equivalent.
B. Query P, Q are not equivalent But if we ignore the duplicates in the result of P, Q then P, Q are equivalent.
C. Query P, Q are not equivalent even if we ignore the duplicates in the result of P, Q.
D. Result of query P is always a superset of the result of query Q.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider an array of $2n$ elements in the form $1, 2n-1, 2, 2n-2, 3, 2n-3, 4, 2n-4, \ldots, n, n$.

For example, here is the array when $n = 8$:

$$1 \quad 15 \quad 2 \quad 14 \quad 3 \quad 13 \quad 4 \quad 12 \quad 5 \quad 11 \quad 6 \quad 10 \quad 7 \quad 9 \quad 8 \quad 8$$

How many compares does Selection Sort algorithm make as a function of n in the worst case? Note that the length of the array is $2n$, not n.

A. $\sim \frac{1}{2}n^2$
B. $\sim n^2$
C. $\sim 2n^2$
D. $\sim 4n^2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

The leftmost column contains an array of 4 integers to be sorted, and the rightmost column contains the integers in sorted order.

Initial Array (A)	Column 1	Column 2	Sorted Array (G)
75	43	34	30
43	75	75	34
34	34	30	43
30	30	43	75

Which of the following is correct regarding sorting algorithms and their intermediate steps?

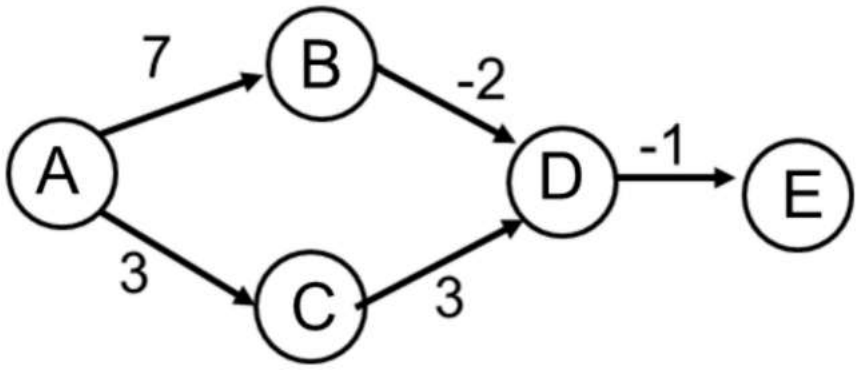
A. Column 1 can be an intermediate step of Mergesort (top-down)
B. Column 2 can be an intermediate step of Quicksort

C. Column 1 can be an intermediate step of Quicksort
D. Column 2 can be an intermediate step of Mergesort (top-down)

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider a weighted graph where we aim to find the shortest paths from source node A to all other nodes using Dijkstra's algorithm.



Which of the following statements is/are correct?

A. The shortest path to node D should be A-B-D, but Dijkstra's algorithm computes A-C-D.
B. The shortest path to node E should be A-B-D-E, but Dijkstra's algorithm computes A-C-D-E.
C. Dijkstra's algorithm correctly computes the shortest path to node D.
D. Dijkstra's algorithm correctly computes the shortest path to node E.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

We found a new implementation of priority queue where insert takes constant time and **extractMin** takes $O(\log n)$ time. Which of the following is true?

A. This can be used to sort in constant time
B. This can be used to sort in linear time
C. This can be used to sort in $O(n \log n)$ time
D. For sorting, one needs a hash table, not a priority queue

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

We have a list L containing non-negative integers. What does the following code in python do?

```
    G=list()
    for i in L:
      if i!=0:
        G.append(i)
5. G += [0]*(len(L)-len(G))
```

A. shifts all non-zero elements to end
B. shifts all zero elements to end
C. shifts non-zero elements by [len(L) - len(G)] places
D. keep only non-zero elements and remove all zeros

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following statements is/are true?

A. For every nonzero vector v in R^3, there exists a basis of R^3 that includes v as an element.
B. Any nonzero vector v in R^2 can be expressed as $v = a + b$, where a and b are nonzero vectors satisfying a $\mathrm{b} = 0$ (orthogonality).
C. If V is a finite-dimensional vector space and $S = \{v_1, v_2, \ldots, v_k\}$ is a spanning set for V , then S contains a subset that forms a basis for V .
D. If x is a solution to $Ax = b$, then x belongs to the row space of A (denoted as Row(A)).

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Suppose A is a 3×4 matrix with the Singular Value Decomposition given as:

$$A = U\Sigma V^T$$

where:
- $U = [u_1, u_2, u_3]$ (orthogonal matrix),
- $\Sigma = \mathrm{diag}(2, 1, 0, 0)$ (diagonal matrix of singular values),
- V is an orthogonal 4×4 matrix.
Which of the following statements is true?

A. The rank of A is 2 , and $\{u_1, u_2\}$ form a basis for the column space of A.
B. The rank of A is 3 , and $\{u_1, u_2, u_3\}$ form a basis for the column space of A.
C. The rank of A is 2 , and $\{v_1, v_2\}$ form a basis for the column space of A.
D. The rank of A is 3 , and $\{v_1, v_2, v_3\}$ form a basis for the column space of A.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Suppose V is an infinite-dimensional vector space, $v_1, v_2, v_3 \in V$ are vectors, and $T : V \to R^2$ is a linear transformation such that:
1. $T(v_1 + 2v_2 + 3v_3) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
2. $T(3v_1 + 2v_2 + v_3) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

What is the value of $T(5v_1 + 2v_2 - v_3)$?

A. $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
B. $\begin{bmatrix} -3 \\ 3 \end{bmatrix}$

C. $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

D. $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the following two statements:

Statement 1 (S1):

If $v_1, v_2, \ldots, v_k \in \mathbb{R}^n$ are linearly independent and W is a subspace of $\mathbb{R}^n$, then the projections $\text{Proj}_W(v_1), \text{Proj}_W(v_2), \ldots, \text{Proj}_W(v_k)$ must also be linearly independent.

Statement 2 (S2):

If W is a subspace of $\mathbb{R}^n$ and x is a vector in $\mathbb{R}^n$ such that $\text{Proj}_W(x) = 0$, then $x \in W^\perp$ (the orthogonal complement of W).

Which of the following is correct?

A. Both S1 and S2 are true.
B. S1 is true, but S2 is false.
C. S1 is false, but S2 is true.
D. Both S1 and S2 are false.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let B be an ordered basis of R^2, consisting of two vectors $v1, \quad v2$. The ordered basis C of R^2 consists of the same vectors in reverse order: v2, v1.

What is the change of basis matrix $\text{P}_{\text{B}\to\text{C}}$?

A. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let V and W be finite-dimensional vector spaces, and let $\text{T} : \text{V} \to \text{W}$ be a linear transformation. Consider the following statements:

A. If $\dim(\text{V}) < \dim(\text{W})$, then T cannot be onto.
B. If $\dim(\text{V}) > \dim(\text{W})$, then T cannot be one-to-one.
C. If $\dim(V) = \dim(W)$, then T is both onto and one-to-one.
D. If $\dim(V) > \dim(W)$, then T must be onto.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X and Y be jointly distributed discrete random variables with the following joint probability mass function:

Y\X	0.3	0.6
0	0.1715	0.032
1	0.2205	0.144
2	0.0945	0.216
3	0.0135	0.108

What is $E[Y \mid X = 0.6]$?

A. 1.2
B. 1.6
C. 1.8
D. 2.0

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Numerical Type | Award: 2 | Penalty: 0 | Probability

A die is rolled twice. Let X equal the sum of the outcomes, and Y equal the first outcome minus the second outcome. What is the value of $\mathrm{Cov}(\mathrm{X}, \mathrm{Y})$?

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

If X and Y are independent and identically distributed random variables with mean μ and variance σ^2, what is the value of $E\left[(X-Y)^2\right]$?

A. 0
B. σ^2
C. $2\sigma^2$
D. $4\sigma^2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let Z be a standard normal random variable such that $P(Z \leq 0.44) = 0.670$.
Let X be a normal random variable with mean 3 and variance 4 .
Find the value of k such that $P(X > k) = 0.33$.

A. $k = 3.44$
B. $k = 3.88$
C. $k = 4.00$
D. $k = 2.56$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

You have ₹1000, and your goal is to maximise your expected money by the end of the week.
There is an opportunity to invest in a commodity. Let k, where $0 \leq k \leq 500$ (since each item costs rupee 2), represent the number of items you buy at the start of the week. At the end of the week, you must sell all k items.
At the beginning of the week, the commodity costs ₹2 per item. The price of the commodity at the end of the week, represented by the random variable R, will be:

- R = ₹1 with probability 0.5

- R = ₹4 with probability 0.5

Let X denote the total amount of money you possess at the end of the week. It is given by: $X = 1000$ - 2k + Rk, where R represents the price of the commodity at the end of the week.

What value of k maximizes $E[X]$, the expected value of X ?

A. $k = 250$, i.e., Buy 250 items initially and sell them all at the end of the week.
B. $k = 500$, i.e., Buy 500 items initially and sell them all at the end of the week.
C. $\mathrm{k} = 0$, i.e., Do not buy any items and keep the₹1000 as cash.
D. $k = 100$, i.e., Buy 100 items initially and sell them all at the end of the week.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let

$$f(x) = \begin{cases} x^2 + x & \text{if } x < 4 \\ c & \text{if } x = 4 \\ 3x + d & \text{if } x \geq 4 \end{cases}$$

Find the values of c and d for which f is continuous at $x = 4$.

A. $c = 20, d = 8$
B. $c = 8, d = 20$
C. $c = 12, d = 8$
D. $c = 20, d = 12$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Which of the following is/are convex functions?

A. $f(x) = x$
B. $f(x) = x^2$
C. $f(x) = e^x$
D. $f(x) = 1/\left(1 + e^{-x}\right)$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Which of the following conditions must hold true (i.e. sufficient condition) for a symmetric matrix A to be positive definite?

A. All eigenvalues are positive (i.e., > 0).
B. All pivots are non-negative (i.e., ≥ 0).
C. The determinant of A is positive.
D. $x^\top Ax > 0$ for all $x \neq 0$.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Assume a finite Taylor series approximation that converges everywhere for a given function $f(x)$. You are given the following information:
- $f(1) = 2$
- $f'(1) = -3$
- $f''(1) = 4$
- $f^{(n)}(1) = 0$ for all $n \geq 3$
Evaluate $f(4)$.

A. 29
B. 11
C. -25
D. -7

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

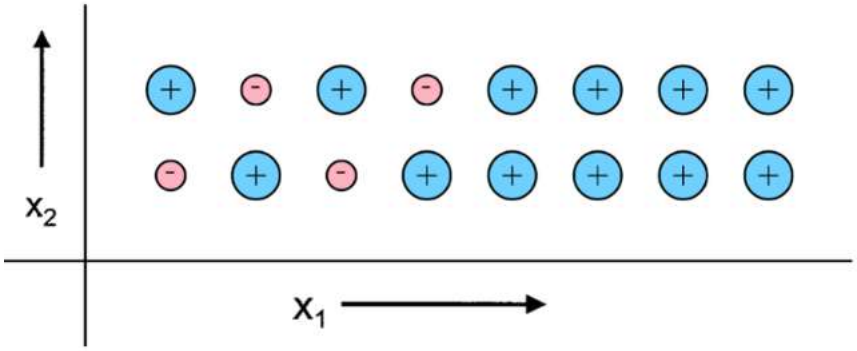
Which of the following statements is/are TRUE regarding bias and variance in linear regression?

A. When conducting linear regression, adding polynomial features to your data often decreases the variance of your fitted model.
B. When conducting linear regression, adding polynomial features to your data often decreases the bias of your fitted model.
C. Suppose your data are an i.i.d. sample from a population. Then collecting a larger sample for use as a training set can help
reduce bias.
D. Suppose your data are an i.i.d. sample from a population. Then collecting a larger sample for use as a training set can help reduce variance.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

The figure below represents a dataset with two inputs, X1 and X2, and one output, Y, which can be either positive (+) or negative (-). The dataset consists of 16 data points: 12 are labeled as positive (+) and 4 are labeled as negative (-).



Assume the algorithm UNDERFIT does not split the data at all, resulting in a decision tree with only one leaf node.
What is the leave-one-out classification error for UNDERFIT on this dataset? Answer the total number of misclassifications.

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We would like to use a perceptron to train a classifier for datasets with 2 features per point and labels +1 or -1 .
Consider the following labeled training data:

Features (x_1, x_2)	**Label y**
(-1, 2)	1
(3, -1)	-1
(1,2)	-1
(3, 1)	1

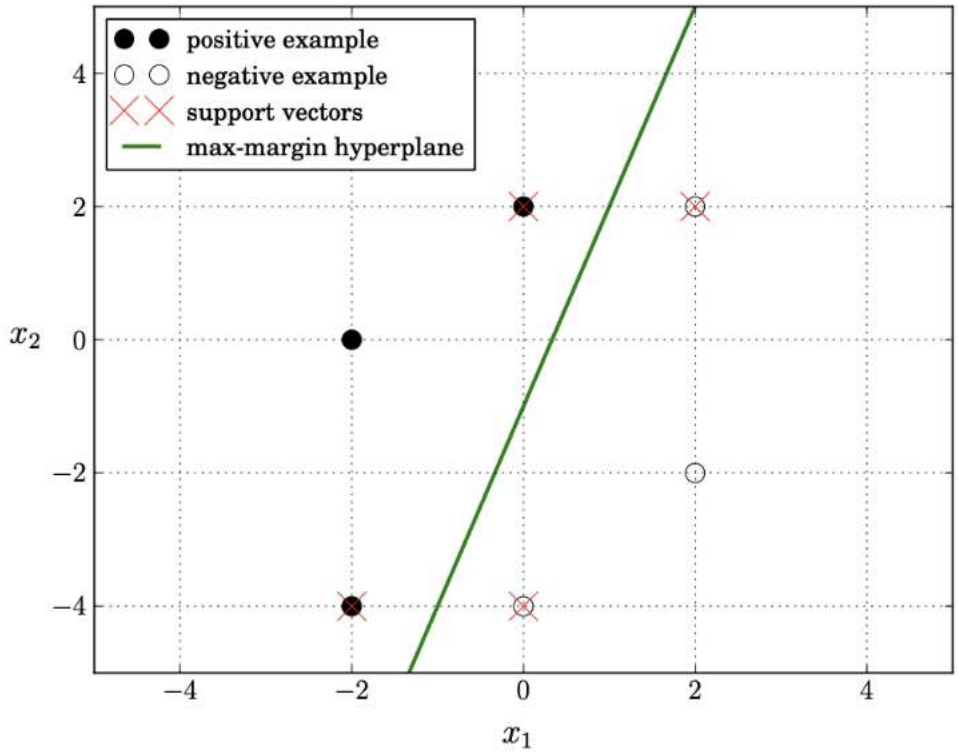
Our perceptron weights are initialized to $\mathrm{w}_1 = 2$ and $\mathrm{w}_2 = -2$. Assume there is no bias. After processing the first point $(-1, 2)$ using the perceptron algorithm, what will be the updated values for these weights?

A. $w_1 = 2, w_2 = -2$
B. $w_1 = 1, w_2 = 0$
C. $w_1 = 3, w_2 = -1$
D. $w_1 = 0, w_2 = -4$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the following 6 training points and the maximum margin classifier drawn in the plot using Hard Margin SVM.



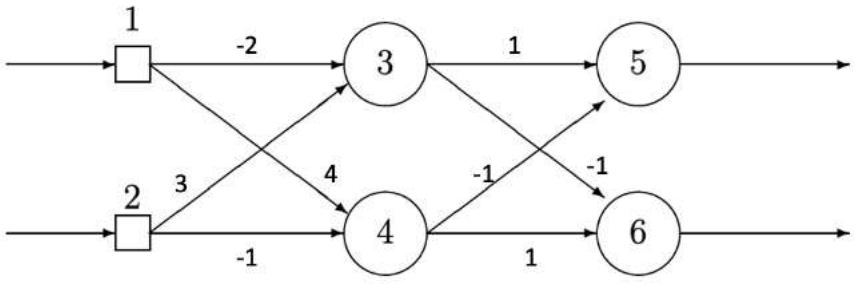
What will the leave-one-out cross-validation error be?

A. 0
B. $1/6$
C. $3/6$
D. $2/6$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

The following diagram represents a feed-forward neural network with one hidden layer. Weights are labelled on each connected in the diagram itself.



Each of the nodes $3, 4, 5$, and 6 uses the activation function:

$$\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

The input pattern P is $(x_1, x_2) = (0, 1)$, What is the output of the network (y_5, y_6) for the input pattern P ?

A. $(1, 0)$
B. $(0, 1)$
C. $(1, 1)$
D. $(0, 0)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

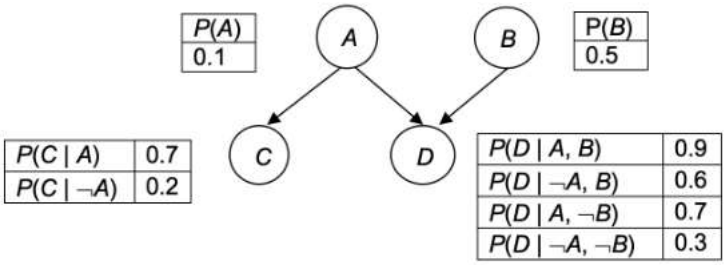
Consider the effects of L_2 regularization in linear regression and the behavior of decision trees in training and testing. Which of the following statements is/are TRUE?

A. In least-squares linear regression, adding L_2 regularization cannot reduce the training L_2 error of the solution, as the unregularized solution already minimizes it.
B. Adding L_2 regularization always decreases the expected L_2 error on unseen test data by preventing overfitting.
C. Decision trees with constant leaves can achieve zero training error for every dataset.
D. In decision tree learning with noise-free data, starting with an incorrect attribute at the root makes it impossible to find a tree that fits the data exactly.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Consider the following Bayesian Network containing four Boolean random variables.



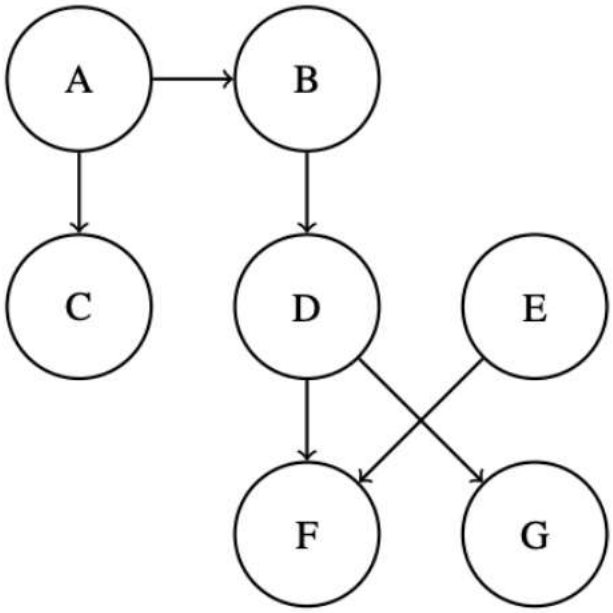
What is $P(A \mid B, C, D)$?

A. 0.0315
B. 0.0855
C. 0.368
D. 0.583

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Which of the following statements is/are true? NOTE : $\mathrm{X} \perp \mathrm{Y}$ represents X is independent of Y and $(X \perp Y \mid Z)$ represents that X is independent of Y given a set of variables Z.



A. F ⊥ G | D
B. B⊥ F | D
C. C⊥G
D. D⊥ E

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

Translate the following English sentence into first-order logic using these predicates:
- Owns(x, y): Object x owns object y.
- $\mathrm{Dog}(x)$: x is a dog.
- Cat(x): x is a cat.
- Cute (x) : x is cute.
- Scary (x) : x is scary.

Sentence: Either Joe has at least one cat and at least one dog, or he is scary (but not both at the same time).

A. $(\exists x($ Owns (Joe, x)$\wedge \mathrm{Dog}(\mathrm{x}))) \wedge (\exists y($ Owns (Joe, y)$\wedge \mathrm{Cat}(\mathrm{y}))) \wedge \neg$ Scary (Joe)
B. $(\exists x(\mathrm{Owns}(\mathrm{Joe}, \mathrm{x}) \wedge \mathrm{Dog}(\mathrm{x}))) \vee (\exists y(\mathrm{Owns}(\mathrm{Joe}, \mathrm{y}) \wedge \mathrm{Cat}(\mathrm{y}))) \wedge \neg$ Scary (Joe)
C. $(\exists x($ Owns $(\mathrm{Joe}, \mathrm{x}) \wedge \mathrm{Dog}(\mathrm{x}))) \wedge (\exists y($ Owns $(Joe, \mathrm{y}) \wedge \mathrm{Cat}(\mathrm{y}))) \Leftrightarrow \neg$ Scary (Joe)
D. $\neg(\exists x(\mathrm{Owns}(\mathrm{Joe}, \mathrm{x}) \wedge \mathrm{Dog}(\mathrm{x}))) \Rightarrow$ Scary(Joe)

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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